

Homework 2: Revision Submission

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Question 1

The distribution of plan counts by county is visualized in Figure 1, where the median of the plan count is shown to increase over time:

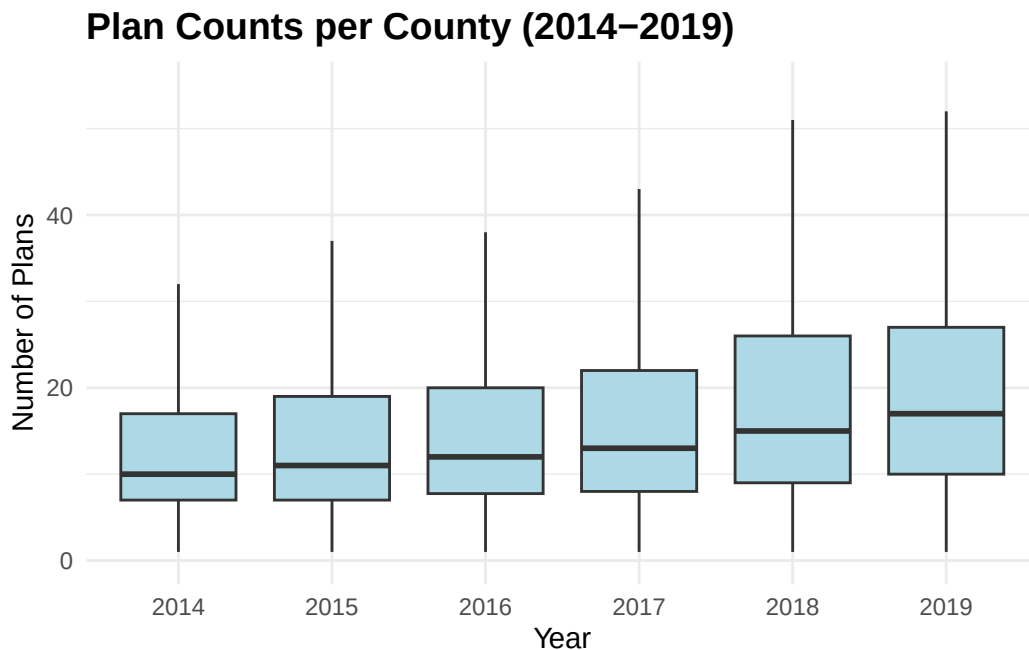


Figure 1: Distribution of Plan Counts per County (2014-2019)

An increase in the median of insurers from 10 insurers per county in 2014 to 17 insurers in 2019 would suggest that the market competition is **sufficient** for beneficiaries to select from a wide variety of plans with various benefits, premiums, and provider networks. More insurers and lower concentration also leads to lower cost-sharing prices and smaller markups that generally favor enrollees.

However, based on the distribution, arguments for beneficiaries in a given county having too many or too few plans can also be justified. For instance, some counties across all six years have access to over 50 plans. Especially for elderly and disabled cohorts that utilize Medicare Advantage (MA), it is often impossible to select from a large catalog of plans that best fits their financial and health circumstances. Beneficiaries are, thus, highly vulnerable to defaulting towards a low-value plan

that is not well-informed, despite the large number of plans ostensibly providing a greater freedom of choice.

Conversely, while this has improved over the years, some counties have less than 10 plans, represented by the lower quartiles of the plots. This may reflect a geographical disparity in the availability of insurers, as beneficiaries residing in rural areas tend to have less access to both health insurance plans and providers compared to urban dwellers. Nevertheless, more data analysis is necessary to support this conclusion.

Question 2

The distribution of MA plan bids is shown in Figure 2:

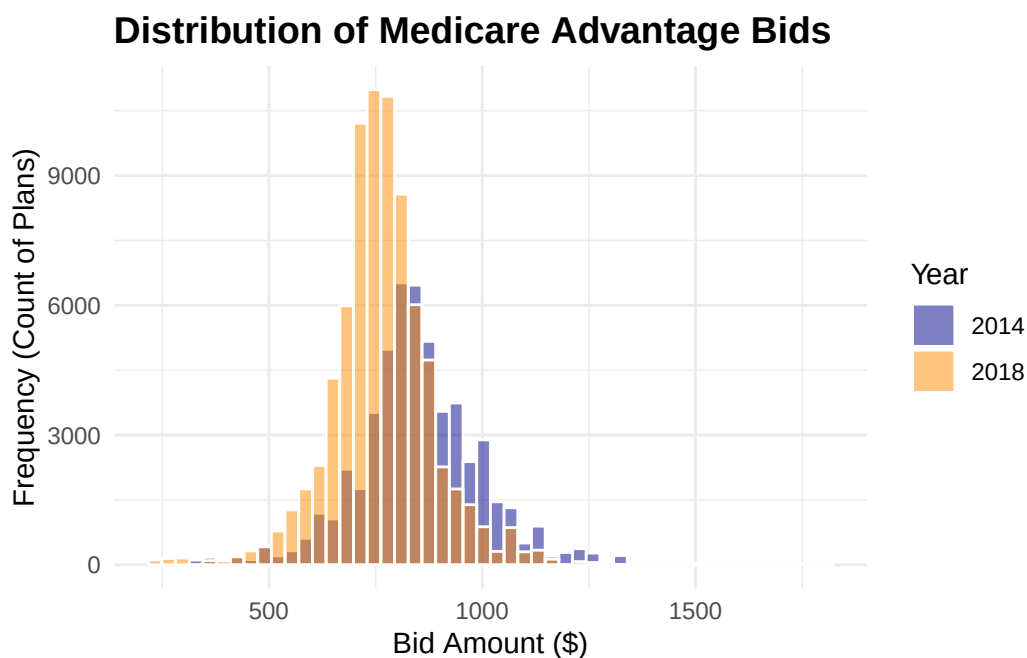


Figure 2: Distribution of Medicare Advantage Bids (2014 vs 2018)

When comparing the distributions between 2014 and 2018, it is remarkably clear that there are more plans in total in 2018 compared to 2014. This supports the previous assertion that the plan count has increased over time.

Another shift to note is that the peak of the distribution shifted slightly to the left in 2018 compared to 2014. This may signal that, as more plans entered the market, insurers felt the pressure of market competition to lower their bids, increasing the difference from the benchmark to offer higher rebates to beneficiaries.

Lastly, the distribution of bids in 2018 is narrower compared to 2014, indicating that the health insurance market may be shifting closer to efficiency as insurers standardize their bids and, subsequently, the benefits they provide to enrollees to stay competitive.

Question 3

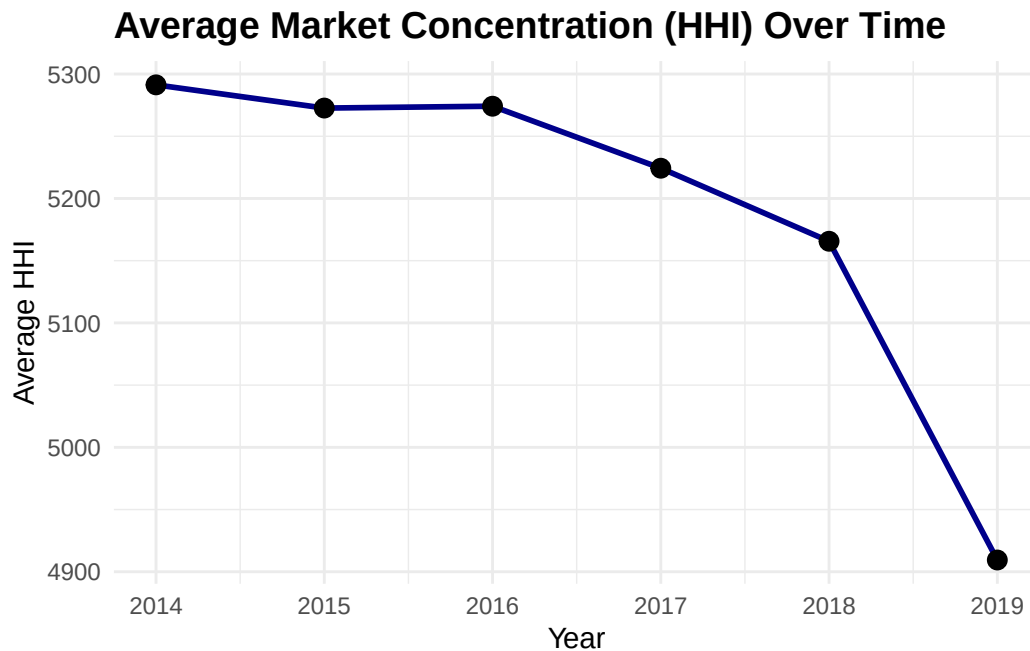


Figure 3: Average Market Concentration (HHI) Over Time (2014-2019)

As mentioned in class, the Herfindahl-Hirschman Index (HHI) measures market concentration, with 0 representing perfect competition and 10,000 representing a complete monopoly.

Based on Figure 3, the average HHI across all counties remained stable at about 5,270 from 2014 to 2016, indicating that the health insurance market was highly concentrated and structured oligopolistically (a few large insurers dominate the industry). The average HHI began to decrease in 2016, with the largest decrease taking place between 2018 (5,166) and 2019 (4,910).

Question 4

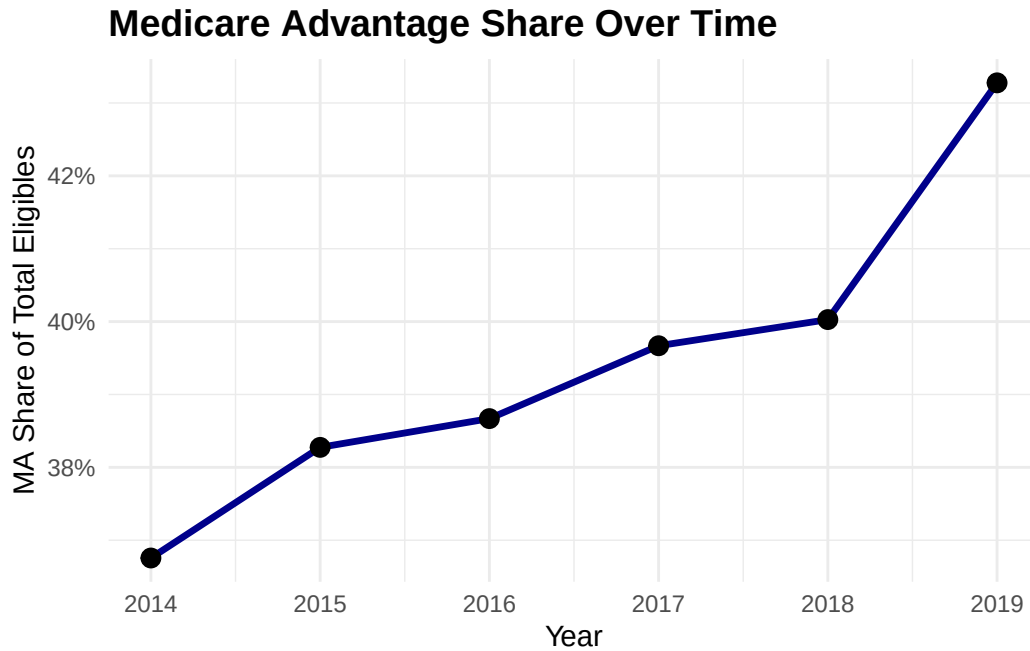


Figure 4: Medicare Advantage Share Over Time (2014-2019)

According to Figure 4, the average share of MA enrollees compared to all Americans that are eligible for Medicare increased from 36.8% in 2014 to 43.3% in 2019.

Question 5

Table 1: Mean Medicare Advantage Bids by HHI (2018)

Market Type	Average Bid
Competitive (Low HHI)	\$732.81
Uncompetitive (High HHI)	\$791.20

Question 6

Table 2: Average Bids by FFS Cost Quartile and Competition Level (2018)

FFS Quartile	Competitive (Low HHI)	Uncompetitive (High HHI)
1	\$777.16	\$788.25
2	\$765.33	\$774.70
3	\$751.50	\$770.97
4	\$744.97	\$783.21

Question 7

Table 3: Table for Question 7: Comparison of ATE Estimators

Estimator	ATE
NN Matching (Inv. Var.)	-\$19.11
NN Matching (Mahalanobis)	-\$18.97
IPW Weighting	-\$19.55
OLS with Interactions	-\$19.55

Question 8

The results for IPW Weighting and simple linear regression with the appropriate interactions are identical in that the average treatment effect (ATE) is -\$19.55, which suggests that, on average, plans in uncompetitive markets submit bids that are \$19.55 less than plans in competitive markets.

While counterintuitive at first glance, this may indicate that insurers in uncompetitive markets have more market power to keep bids low in order to maximize rebates, attracting more beneficiaries in the process.

Table 2 tells a different story, where the raw averages indicate that plans in competitive markets tend to submit lower bids compared to plans in uncompetitive markets. However, this is likely due to selection bias. For instance, if uncompetitive counties consistently have higher, underlying medical costs or fewer beneficiaries than competitive counties, then average bids for uncompetitive counties will be higher.

Question 9

Table 4: Sensitivity Analysis: Quartile vs. Continuous Controls

Model Spec	ATE (\$)
FFS Cost Quartiles (Q7)	-19.55
Continuous FFS Costs & Eligibles (Q9)	-16.21

Question 10

Overall, I thought that the process of visualizing data was very fun. Seeing the plots being rendered through code when I typically use default plots from excel was delightful and I'm excited to see what the other homeworks have in store for me in the future. One thing I learned, conceptually, was HHI. I've heard about it for the first time in class, where I later learned about market concentration and how that affects competition as well. Seeing that HHI decreases as the number of insurers increases by year from question 3 helped to create a more comprehensive story with the data. One thing that surprised me that, frankly, I'm still trying to figure out is the ATE estimators from question 7 showing that uncompetitive markets submit lower bids, on average, than competitive markets. Perhaps this is due to efficiency or increased market power from having few firms, but I expected higher bids in uncompetitive markets since fewer firms, paradoxically, have higher market power to inflate prices.